Adult Census Income Fairness Analysis — Computations

Dataset: UCI Adult Census Income. The target is 1 for income above $50K and 0 for income of $50K or less. Random state: 156. The data was split into 70% for training and 30% for testing.

1. Dataset and Preprocessing

The dataset had 32,561 rows and 15 columns. Values marked with a question mark were treated as missing. There were 1,836 missing values in workclass, 1,843 in occupation, and 583 in native-country. These missing values were filled using the most common value in each column. Categorical columns were converted into numeric columns using one-hot encoding. The numeric columns age, fnlwgt, education-num, capital-gain, capital-loss, and hours-per-week were scaled. The scaler was fitted on the training data and then used on the test data. Extreme values were kept; no separate outlier-removal step was used.

Dataset item	Result
Original dataset	32,561 rows × 15 columns
Training features	22,792 rows × 97 features
Test features	9,769 rows × 97 features
Target	Income > $50K = 1; income ≤ $50K = 0
Train-test split	70% training / 30% testing
random_state	156

2. Model A and Model B

Model A uses Linear Regression (OLS). Its predictions were split into two classes using the median prediction as the cut-off. Model B uses Logistic Regression, with 0.50 as the cut-off probability.

Calculations: OLS class = 1 when its prediction is at least the median; otherwise 0. Logistic class = 1 when its probability is at least 0.50; otherwise 0. Mismatch percentage = number of different predictions ÷ test-set size × 100.

Measure	Result
OLS median cut-off	0.2285
Logistic Regression cut-off	0.5000
Test-set size	9,769
Different predictions	2,978
Mismatch percentage	30.48%

3. Logistic Regression by Sex

The model was fitted to all training data and then fitted separately for the male and female groups. The table shows the results for four selected numeric features.

Feature	Full dataset	Male group	Female group
age	0.2931	0.2751	0.3772
education-num	0.8136	0.7557	1.0530
hours-per-week	0.3706	0.3842	0.3043
capital-gain	2.2285	2.1245	2.5453

For education-num, the female coefficient is 1.0530 and the male coefficient is 0.7557. The difference is 1.0530 − 0.7557 = 0.2973. The results do not show a change in coefficient sign. Confidence intervals, p-values, and an interaction test were not included in the notebook results.

4. Omitted-Variable Bias

The model was run again after removing education-num. The change is the reduced-model coefficient minus the full-model coefficient. Percentage change = (reduced coefficient − full coefficient) ÷ full coefficient × 100.

Feature	Full model	Reduced model	Change	Change (%)
age	0.2931	0.2865	−0.0065	−2.2287%
hours-per-week	0.3706	0.3707	+0.0000*	+0.0125%
capital-gain	2.2285	2.2337	+0.0052	+0.2343%
marital-status_Married-civ-spouse	1.8238	1.7486	−0.0752	−4.1256%

*The hours-per-week change rounds to 0.0000. Its percentage change, calculated from the unrounded values, is 0.0125%.

5. High-Confidence Wrong Predictions

A high-confidence wrong prediction means the model gave a probability of at least 0.75 when the true answer was 0, or a probability of at most 0.25 when the true answer was 1. The notebook found 477 such cases. This is 477 ÷ 9,769 × 100 = 4.88% of the test set. A sample of 50 cases was used for the comparison below.

Measure	Sample of wrong predictions	Full test set
Number of rows	50	9,769
Average age	43.48	38.70
Female share	20.00%	33.31%
Male share	80.00%	66.69%
Average education-num	10.36	10.10

6. Fairness Results at Different Cut-offs

The groups are male and female. At each cut-off, a prediction was counted as positive when its probability was at least that value. The table shows the DP, EO, and PP ratios calculated by the notebook.

Cut-off	DP	EO	PP
0.05	2.1257	1.0444	1.2797
0.15	3.1353	1.1481	0.9538
0.25	3.3750	1.2144	0.9373
0.35	3.3966	1.2562	0.9633
0.45	3.3935	1.2138	0.9317
0.55	3.3112	1.1802	0.9284
0.65	3.3805	1.2313	0.9488
0.75	3.1547	1.1985	0.9895
0.85	3.0301	1.1728	1.0081
0.95	2.9833	0.9556	1.0428

7. Notes

The notebook calculates the mismatch rate as 30.48% and the high-confidence wrong-prediction rate as 4.88%. The fairness values above are copied from the notebook's calculations. Its EO calculation uses ratios for the two groups and can give values above 1.0, so check that calculation against the definition required by the assignment before interpreting it.